

EMPLOYEE RAISE ISSUE – RECORD PRODUCER & SERVICE PORTAL INTEGRATION

A ServiceNow-Based Employee Issue Management System

TABLE OF CONTENTS

- 1. Ideation Phase**
 - 1.1 Project Idea
 - 1.2 Problem Statement
 - 1.3 Proposed Solution
 - 1.4 Objectives
 - 1.5 Scope
- 2. Requirement Analysis Phase**
 - 2.1 Functional Requirements
 - 2.2 Non-Functional Requirements
 - 2.3 Software Requirements
 - 2.4 Hardware Requirements
- 3. Project Design Phase**
 - 3.1 System Architecture
 - 3.2 Application Design
 - 3.3 Custom Table Design
 - 3.4 Application Components
 - 3.5 Application Manager View
 - 3.6 UI Policy Design
 - 3.7 Record Producer Design
 - 3.8 Service Portal Design
- 4. Project Planning Phase**
 - 4.1 Project Development Plan
 - 4.2 Time Planning
 - 4.3 Testing Plan
 - 4.4 Development Planning
 - 4.5 Portal and Widget Planning
 - 4.6 Integration Planning
 - 4.7 Expected Outcome
- 5. Project Development Phase**
 - 5.1 Creating the Scoped Application
 - 5.2 Creating the Custom Table
 - 5.3 Configuring Fields
 - 5.4 Implementing UI Policies
 - 5.5 Creating the Record Producer
 - 5.6 Creating the Commercial Services Widget
 - 5.7 Creating the Raise an Issue Widget

- 5.8 Configuring the Service Portal Homepage
- 5.9 Configuring the Raise Employee Issue Form
- 5.10 Submitting an Employee Issue
- 5.11 Verifying Backend Record Creation

6. System Workflow

7. Testing and Validation

- 7.1 Portal Access Test
- 7.2 Raise Employee Issue Navigation Test
- 7.3 Form Submission Test
- 7.4 Record Producer Test
- 7.5 UI Policy Validation
- 7.6 End to End Flow Validation

8. Project Outcome

9. Benefits of the System

10. Future Enhancements

11. Conclusion

1. IDEATION PHASE

1.1 Project Idea

The project aims to develop an Employee Raise Issue System using ServiceNow. The system provides employees with a simple and centralized way to report workplace issues through a Service Portal.

Instead of depending on emails, phone calls, or manual communication with HR or the concerned department, employees can submit their issues through an online form. The submitted information is stored in ServiceNow as an issue record.

The project uses ServiceNow components such as a custom scoped application, custom table, UI Policies, Record Producer, Service Portal, and custom widgets.

1.2 Problem Statement

In a traditional organization, employees may face difficulties when reporting workplace issues because there may not be a centralized issue-reporting system.

The major problems include:

- Employees may need to contact HR or other departments manually.
- Tracking the status of an issue can be difficult.
- Important issue information may be lost in emails or manual records.
- There is limited visibility into previously submitted issues.
- Manual issue reporting can take additional time.
- There may not be a simple employee-facing interface for submitting issues.

Therefore, a centralized ServiceNow-based system is proposed to simplify the issue-reporting process.

1.3 Proposed Solution

The proposed solution is a ServiceNow-based Employee Raise Issue System.

The employee can:

1. Open the Employee Service Portal.
2. Select the Raise Employee Issue option.
3. Enter the required issue details.
4. Submit the form.
5. The Record Producer processes the submitted information.
6. A new record is automatically created in the custom Employee Issue table.
7. The record receives a unique issue number.

The overall solution can be represented as:

Employee → Service Portal → Raise Employee Issue → Record Producer → Custom Table → Employee Issue Record

1.4 Objectives

The main objectives of the project are:

- To provide an easy-to-use employee issue reporting system.
- To centralize employee issue information.
- To reduce manual issue reporting.
- To provide a structured issue submission form.
- To automate record creation using a Record Producer.
- To provide a user-friendly Service Portal interface.
- To organize submitted employee issues in a custom ServiceNow table.
- To provide a foundation for future automation.

1.5 Scope of the project

The current scope of the project includes:

- Creating a custom scoped ServiceNow application.
- Creating a custom Employee Issue table.
- Creating and configuring required fields.
- Implementing UI Policies.
- Creating a Record Producer.
- Configuring the Service Portal.
- Creating custom Service Portal widgets.
- Creating the Raise Employee Issue interface.
- Connecting the employee-facing form with the backend table.
- Testing issue submission and record creation.

2. REQUIREMENT ANALYSIS PHASE

2.1 Functional Requirements

Employee

The employee should be able to:

- Access the Service Portal.
- Open the Raise Employee Issue form.
- Enter issue details.
- View required fields.
- Submit an issue.
- Create a new issue record through the Record Producer.

ServiceNow Administrator

The administrator should be able to:

- View submitted employee issues.
- Manage issue records.
- Configure application components.
- Configure table fields.
- Maintain UI Policies.
- Maintain the Record Producer.
- Configure the Service Portal and widgets.

2.2 Non-Functional Requirements

Usability

The portal and form should be simple and easy for employees to use.

Security

Only authorized users should be able to access and manage appropriate records.

Reliability

Submitted employee issues should be stored correctly without data loss.

Maintainability

The application should be easy to modify and extend in the future.

Performance

The issue submission process should respond within a reasonable time.

2.3 Software Requirements

- ServiceNow Personal Developer Instance (PDI)
- ServiceNow Service Portal
- ServiceNow Studio/Application tools

- Web Browser
- ServiceNow Record Producer
- ServiceNow UI Policies
- ServiceNow Service Portal Designer

2.4 Hardware Requirements

- Computer/Laptop
- Internet connection
- Minimum 4 GB RAM recommended
- Modern web browser

3. PROJECT DESIGN PHASE

3.1 System Architecture

The system architecture represents the flow of an employee issue from the user interface to the backend record. The employee accesses the Service Portal and selects the **Raise an Issue** option. The **Raise an Issue Widget** provides navigation to the **Raise Employee Issue Record Producer**. The Record Producer collects the employee's issue details and creates a record in the **Employee raises issue** custom table. **UI Policies** control the behavior of fields on the form based on the configured conditions.

Employee → Service Portal → Raise an Issue Widget → Raise Employee Issue Record Producer → UI Policies → Employee raises issue Table → Employee Issue Record

3.2 Application Design

A custom scoped application was created to manage employee issues.

The major components are:

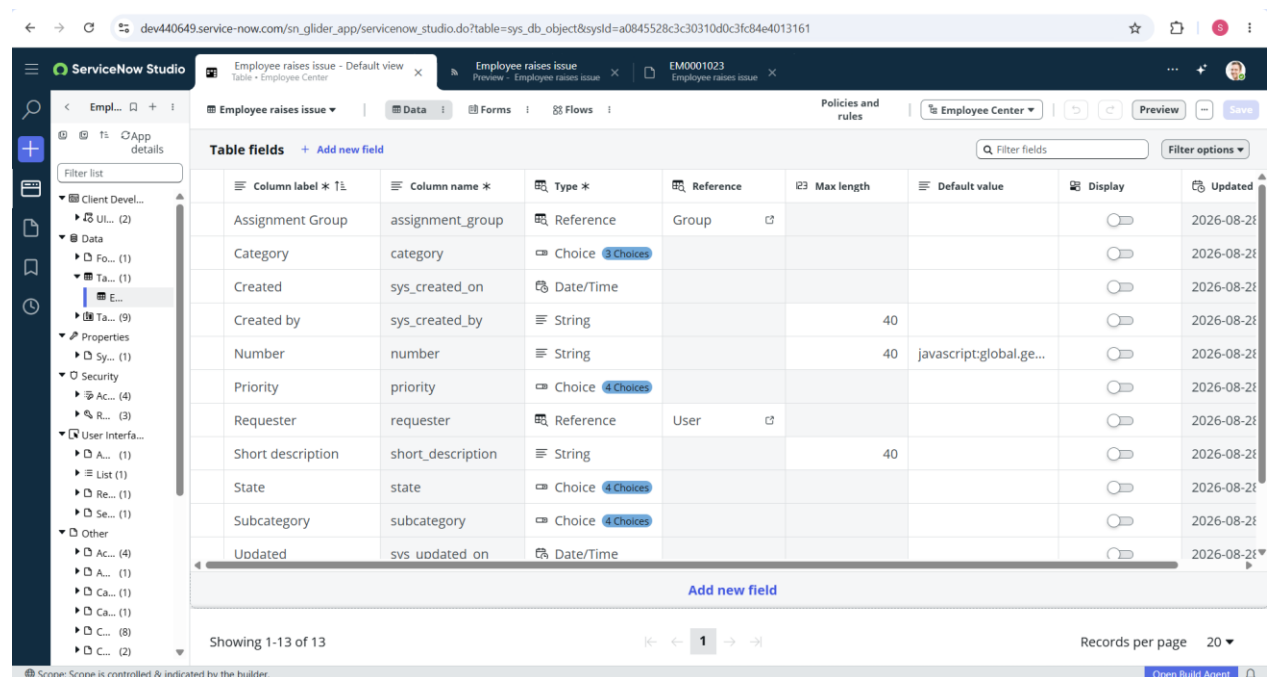
- Custom Scoped Application
- Custom Employee Issue Table
- Custom Fields
- UI Policies
- Record Producer
- Service Portal
- Custom Widgets
- Raise Employee Issue Form

3.3 Custom Table Design

A custom table was created to store employee issue information. The table acts as the backend data source for the application.

The implemented table contains fields such as:

- Assignment Group
- Category
- Created
- Created By
- Number
- Priority
- Requester
- Short Description
- State
- Subcategory



The screenshot displays the ServiceNow Studio interface for a custom table named 'Employee raises issue'. The table is configured with the following fields:

Column label *	Column name *	Type *	Reference	Max length	Default value	Display	Updated
Assignment Group	assignment_group	Reference	Group			☐	2026-08-28
Category	category	Choice (3 Choices)				☐	2026-08-28
Created	sys_created_on	Date/Time				☐	2026-08-28
Created by	sys_created_by	String		40		☐	2026-08-28
Number	number	String		40	javascript:global.ge...	☐	2026-08-28
Priority	priority	Choice (4 Choices)				☐	2026-08-28
Requester	requester	Reference	User			☐	2026-08-28
Short description	short_description	String		40		☐	2026-08-28
State	state	Choice (4 Choices)				☐	2026-08-28
Subcategory	subcategory	Choice (4 Choices)				☐	2026-08-28
Updated	sys_updated_on	Date/Time				☐	2026-08-28

The interface includes a left-hand navigation pane with various tool categories, a top toolbar with tabs for 'Data', 'Forms', and 'Flows', and a bottom status bar indicating 'Showing 1-13 of 13' records and 'Records per page 20'.

Figure 1. Employee Issue Custom Table Fields in ServiceNow Studio

3.4 Application Components

The ServiceNow Studio application view shows the custom application and its configured application components, including the custom table and Record Producer.

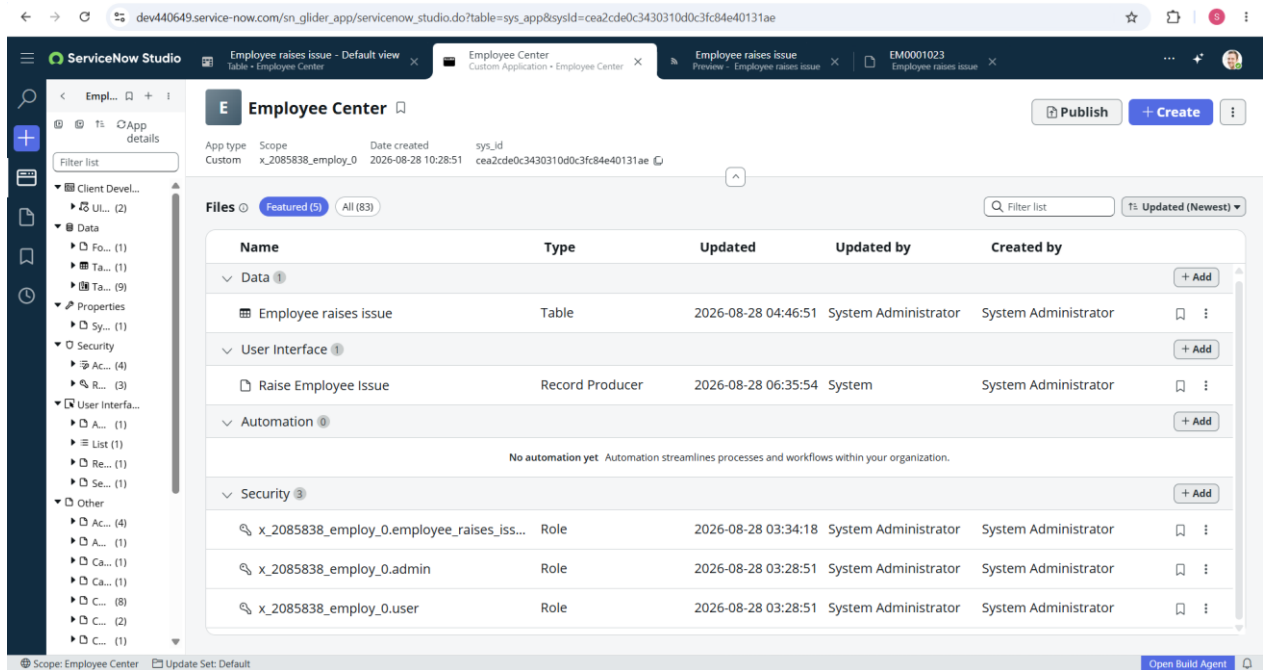


Figure 2. Custom Scoped Application and Configured Application Components

3.5 UI Policy Design

UI Policies are used to control the behavior of fields on the form.

They can be used to make fields mandatory, visible/hidden, or read-only based on defined conditions.

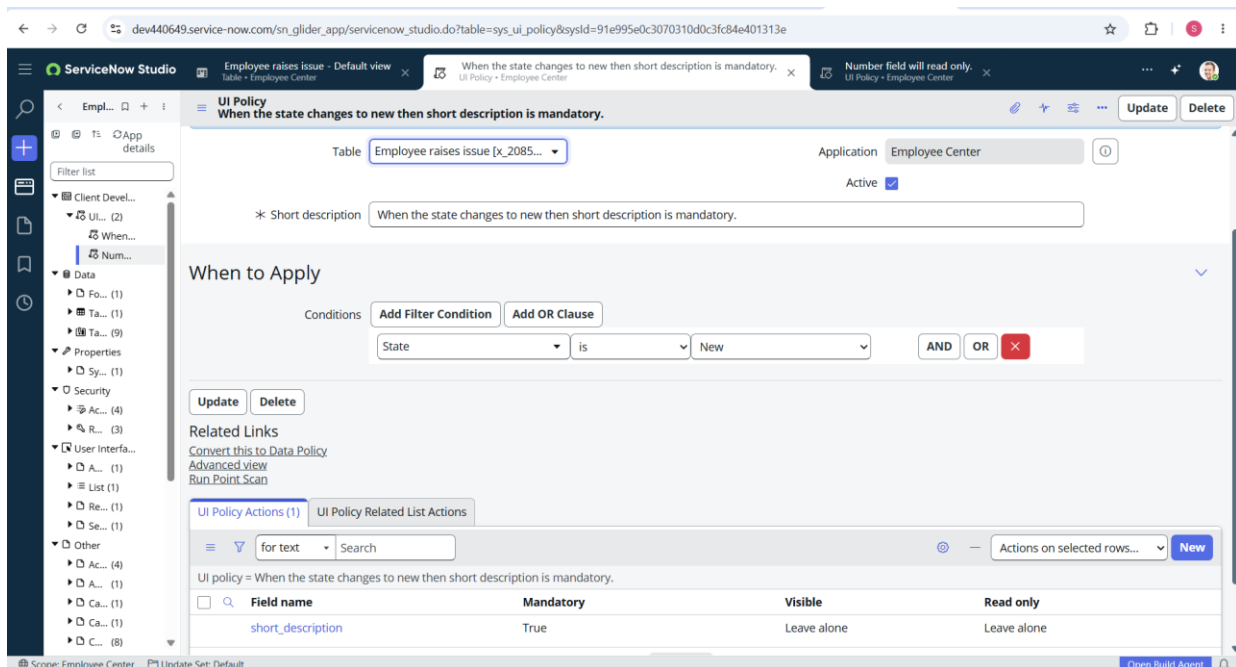


Figure 3. UI Policy Configuration for the Employee Issue Form

3.6 Record Producer Design

The Record Producer provides an employee-friendly form for submitting an issue.

The process is:

Employee opens form → Enters details → Submits form → Record Producer processes information → Custom table record is created

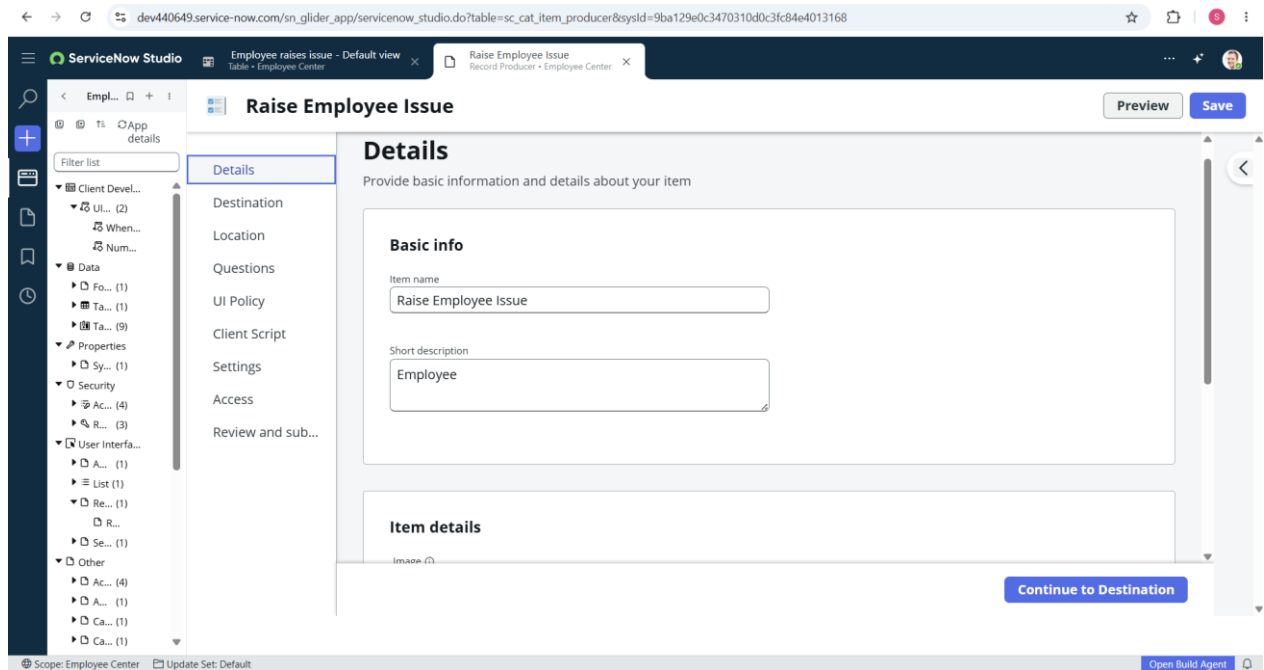


Figure 4. Raise Employee Issue Record Producer Configuration

3.7 Service Portal Design

A Service Portal was configured to provide the user interface for employees.

The portal contains:

- Homepage
- Employee issue service
- Raise Employee Issue option
- Navigation/interface for accessing the service

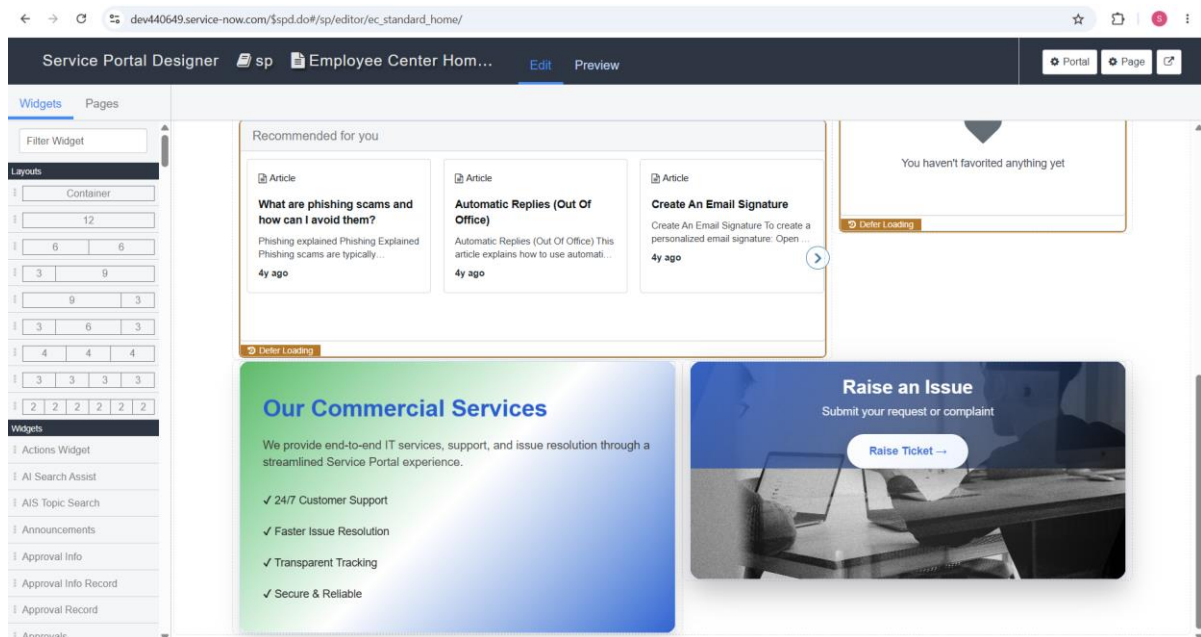


Figure 5. Service Portal Designer Showing the Employee Portal Pages

4. PROJECT PLANNING PHASE

4.1 Project Development Plan

The project was planned in sequential stages.

First, the problem and project objectives were identified. The requirements were then analyzed before beginning the ServiceNow implementation.

The development activities included creating the custom scoped application, creating the custom table, configuring fields, implementing UI Policies, creating the Record Producer, configuring the Service Portal, developing widgets, and creating the Raise Employee Issue form.

After development, the complete workflow was tested from the employee portal to backend record creation.

The planned development sequence was:

Problem Identification → Requirement Analysis → Application Creation → Table Creation → Field Configuration → UI Policies → Record Producer → Service Portal → Widget Development → Form Testing → Record Creation Testing → Documentation

4.2 Time Planning

The project was planned to be completed in different stages, starting with requirement identification and design, followed by application development, portal configuration, widget development, integration, and testing.

Each stage was completed before moving to the next stage to reduce configuration errors and ensure that the system worked correctly.

4.3 Testing Plan

The system was planned to be tested from the employee's perspective.

The main testing activities included:

- Opening the Employee Service Portal.
- Accessing the Raise Employee Issue option.
- Opening the issue form.
- Entering issue details.
- Submitting the form.
- Verifying Record Producer processing.
- Checking whether the record was created in the custom table.
- Verifying the generated issue number.

4.4 Development Planning

The development was planned around the main ServiceNow components.

The custom application provides the project structure, while the custom table provides backend storage. UI Policies provide form behavior, and the Record Producer provides the employee-facing issue submission mechanism.

4.5 Portal and Widget Planning

The Service Portal was planned as the main employee-facing interface.

Custom widgets were planned to provide clear navigation and an easy-to-use interface for accessing the Raise Employee Issue functionality.

4.6 Integration Planning

The portal, Record Producer, and custom table were planned to work together as a single workflow.

The employee enters information through the portal. The Record Producer processes the information and creates the corresponding record in the backend custom table.

4.7 Expected Outcome

The expected outcome was a functional ServiceNow-based Employee Raise Issue System where employees can easily report issues through a centralized portal.

The system should provide a simple interface for submitting issues and automatically create corresponding records for further management.

5. PROJECT DEVELOPMENT PHASE

5.1 Creating the Scoped Application

A custom scoped application was created in ServiceNow to keep the project components organized within a dedicated application.

5.2 Creating the Custom Table

A custom table named Employee raises issue was created to store employee issue records. The table acts as the backend data source for the application.

5.3 Configuring Fields

Fields were added and configured to capture employee and issue-related information. The fields provide structured data that can be stored and managed in the custom table.

5.4 Implementing UI Policies

UI Policies were created to control form behavior and ensure that important information is properly collected.

5.5 Creating the Record Producer

A Record Producer was created to allow employees to create issue records through a simplified Service Portal form.

The Record Producer maps the information entered by the employee to the appropriate fields in the custom Employee Issue table.

5.6 Creating the Commercial Services Widget

A custom Service Portal widget was developed to present the employee-facing service information in a visually organized manner.

The widget provides the portal user with service-related information and helps make the portal interface more user-friendly.

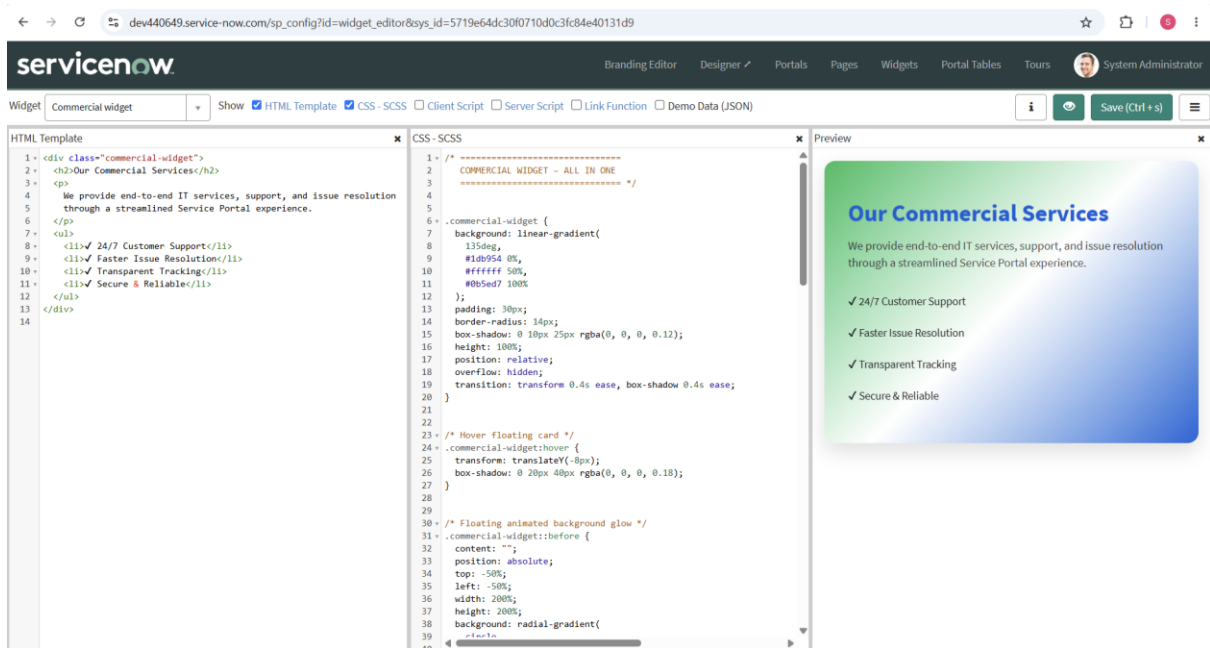


Figure 6. Custom Commercial Services Widget

5.7 Creating the Raise an Issue Widget

A custom Raise an Issue widget was developed to provide a direct entry point to the issue submission functionality.

The widget provides navigation from the Service Portal homepage to the Raise Employee Issue functionality.

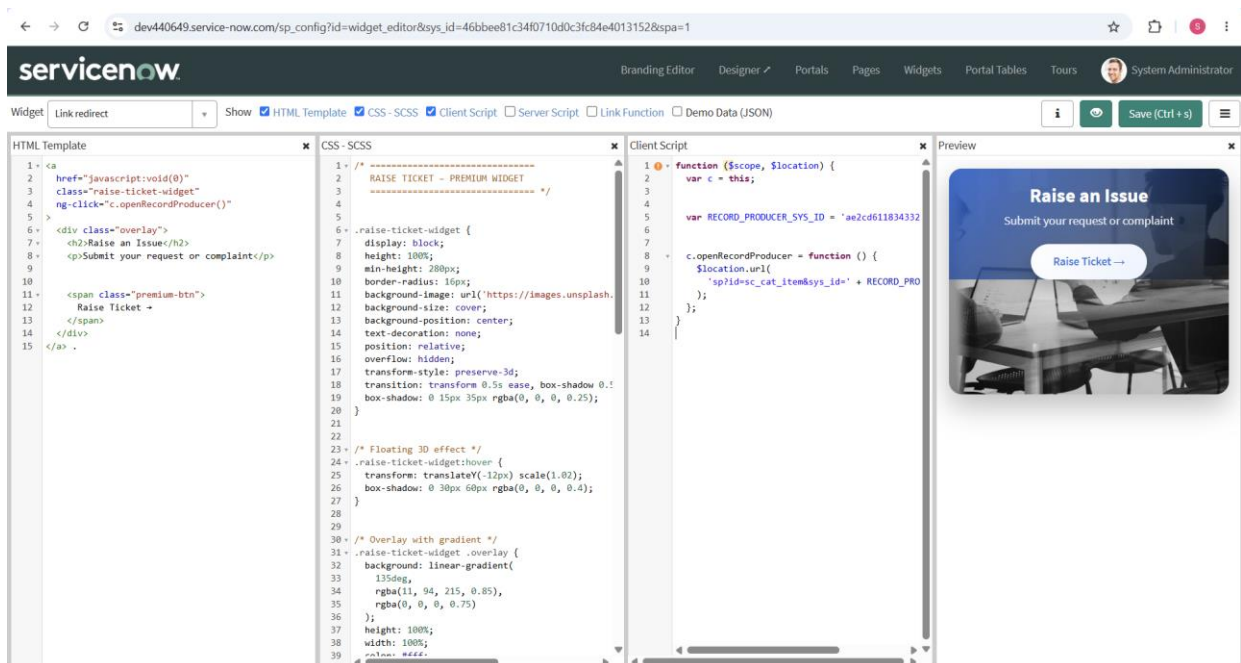


Figure 7. Custom Raise an Issue Widget and Navigation Configuration

5.8 Configuring the Service Portal Homepage

The custom widgets were placed on the Service Portal homepage using Service Portal Designer.

The homepage was configured to provide employees with an easy way to access the issue-reporting functionality.

5.9 Configuring the Raise Employee Issue Form

The Raise Employee Issue form provides the employee with an interface for entering the required issue information.

The employee can enter the relevant details and submit the form through the portal.

5.10 Submitting an Employee Issue

After entering the required information, the employee submits the issue.

The Record Producer processes the submitted information and creates a corresponding record in the custom Employee Issue table.

5.11 Verifying Backend Record Creation

After successful submission, the generated record can be verified in the custom Employee Issue table.

The system generates a unique issue number for the submitted record.

For example:

The screenshot displays the ServiceNow interface for creating a new employee issue. The browser address bar shows the URL: `dev440649.service-now.com/now/nav/ui/classic/params/target/x_2085838_employ_0_employee_raises_issue.do%3Fsys_id%3Dddfd3c1c3834710d0c3fc84e40131f5%26sysparm_record_ta...`. The ServiceNow header includes the logo, navigation tabs (All, Favorites, History, Workspaces), and a search bar. The main title is "Employee raises issue - EM0001022". Below the title, there are "Update" and "Delete" buttons. The form fields are as follows:

Number	EM0001022
Requester	Aileen Mottern
Category	Access
Subcategory	forgot password
* Short description	password access
State	New
Priority	-- None --
Assignment Group	

At the bottom of the form, there are "Update" and "Delete" buttons.

Figure 8. Backend Record Creation Form

6. SYSTEM WORKFLOW

The complete implemented workflow is:

Employee



Open Employee Service Portal



Select Raise Employee Issue



Raise Employee Issue Form



Enter Required Details



Submit Form



Record Producer



Custom Employee Issue Table



Employee Issue Record Created



Unique Issue Number Generated

The workflow demonstrates the connection between the employee-facing Service Portal and the backend ServiceNow application.

7. TESTING AND VALIDATION

7.1 Portal Access Test

Input: Employee opens the Employee Service Portal.

Expected Result: The portal homepage should load successfully and display the configured Commercial Services and Raise an Issue widgets.

Result: Passed.

7.2 Raise Employee Issue Navigation Test

Input: Employee selects the **Raise an Issue** option from the Service Portal.

Expected Result: The **Raise Employee Issue** form should open successfully.

Result: Passed.

7.3 Form Submission Test

Input: Employee enters the required issue information and submits the form.

Expected Result: The issue should be submitted successfully through the Record Producer.

Result: Passed

7.4 Record Producer Test

Input: A valid issue is submitted through the Raise Employee Issue form.

Expected Result: A new record should be created in the **Employee raises issue** custom table with a unique issue number and the submitted information.

Result: Passed.

7.5 UI Policy Validation

Input: Employee opens the issue form and enters the required information.

Expected Result: The configured UI Policy should apply the required field behavior correctly.

Result: Passed.

7.6 End-to-End Flow Validation

Input: Employee accesses the portal, selects Raise an Issue, enters issue details, and submits the form.

Expected Result: The request should move successfully through the complete flow:

Employee → Service Portal → Raise an Issue → Record Producer → Submit → Employee raises issue Record

Result: Passed.

7.7 Testing Result

The testing confirmed that the main implemented functionality of the Employee Raise Issue System works successfully. The Service Portal loads correctly, the Raise Employee Issue form opens through the configured widget, issue details can be submitted, and the submitted information is stored as a new record in the custom table. The UI Policy behavior was also verified successfully. This confirms that the employee-facing portal and backend record creation are properly connected.

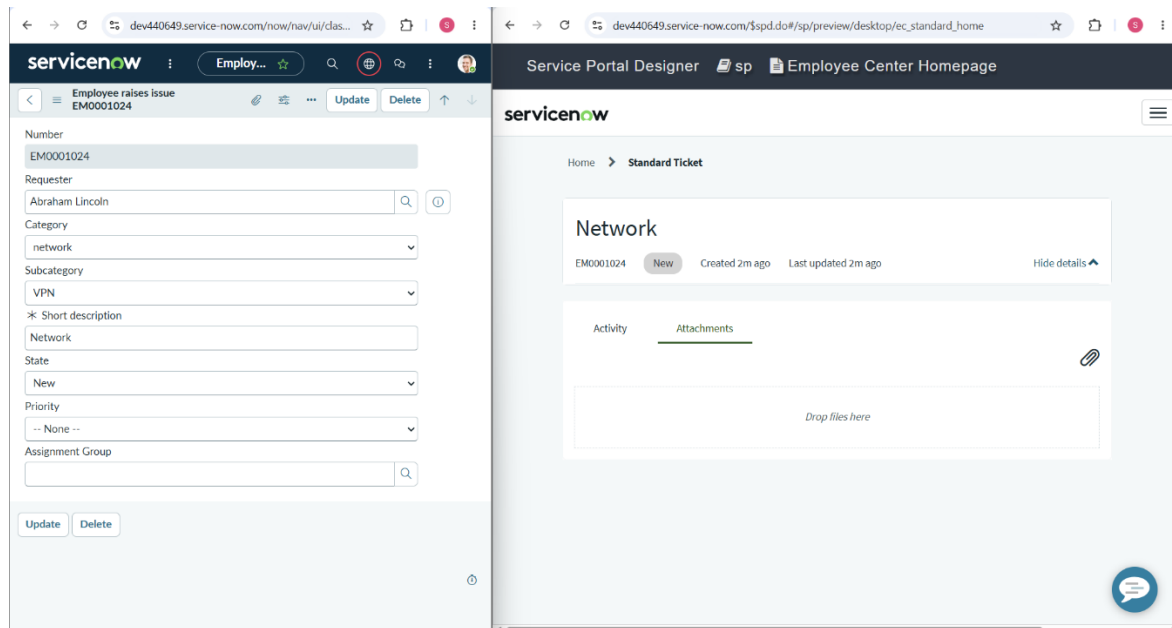


Figure 9. Successful Employee Issue Record Creation and Service Portal Ticket View

8. PROJECT OUTCOME

The project provides a centralized ServiceNow-based method for employees to report workplace issues.

The Service Portal acts as the user-facing interface, while the Record Producer simplifies issue submission and creates records in the custom Employee Issue table.

The implemented system successfully provides:

- Custom Employee application.
- Employee Issue custom table.
- Required fields and form configuration.
- UI Policies.
- Raise Employee Issue Record Producer.
- Service Portal.
- Custom Commercial Services widget.
- Custom Raise an Issue widget.
- Customized portal homepage.
- Employee issue submission.
- Backend record creation.
- Unique issue number generation.

9. BENEFITS OF THE SYSTEM

The Employee Raise Issue System provides several benefits:

Centralized Issue Management

Employee issues are stored in a centralized ServiceNow table.

Easy Issue Submission

Employees can report issues through a simple Service Portal interface.

Reduced Manual Communication

The system reduces dependence on manual reporting methods such as emails and direct communication.

Structured Information

Issue information is captured through predefined fields.

Automatic Record Creation

The Record Producer automatically creates the corresponding backend record.

User-Friendly Interface

The Service Portal provides a simple interface for employees.

Extensibility

The ServiceNow application can be enhanced with additional automation and management features in the future.

10. FUTURE ENHANCEMENTS

The current system can be extended with the following features:

- Automated email notifications.
- Approval workflow.
- Automatic issue assignment to departments.
- Issue status tracking for employees.
- Employee notifications when status changes.
- SLA management.
- Dashboard and reporting.
- Automated escalation for high-priority issues.
- Integration with HR systems.

These are future enhancements, not currently implemented features. This distinction should be maintained in the final report.

11. CONCLUSION

The Employee Raise Issue – Record Producer & Service Portal Integration project provides a centralized and user-friendly solution for employees to report workplace issues using ServiceNow.

The Service Portal provides the front-end interface, while the Record Producer simplifies issue submission and automatically creates records in the custom Employee Issue table.

The project successfully demonstrates the integration of a custom scoped application, custom table, configured fields, UI Policies, Record Producer, Service Portal, and custom widgets.

The implemented workflow successfully connects the employee-facing portal with the backend ServiceNow table and verifies the creation of employee issue records with unique issue numbers.

The system also provides a foundation for future enhancements such as notifications, approvals, SLA management, dashboards, automated assignment, escalation, and HR-system integration.